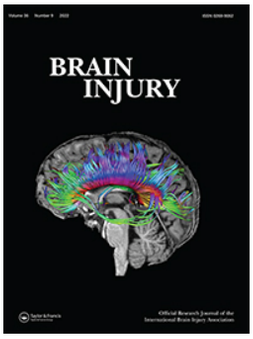


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# Neurocognitive outcomes of older National Football League retirees

Jeff Schaffert[a], Nyaz Didehbani[a,b], Christian LoBue[a,c], John Hart Jr[a,d,e], Michael Motes[d], Heidi Rossetti[a,e], Kristin Wilmoth[b], Will Goette[a], Laura Lacritz[a,e], and C. Munro Cullum[a,c,e]

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**ABSTRACT**

**Objective:** Determine if head-injury exposure relates to later-in-life cognitive decline in older National Football League (NFL) retirees.

**Method:** NFL retirees (aged 50+) with or without cognitive impairment underwent baseline (n = 53) and follow-up (n = 29; 13–59 months later) neuropsychological evaluations. Cognitively normal (CN) retirees (n = 26) were age- and education-matched to healthy controls (n = 26). Cognitively impaired (CI) retirees with mild cognitive impairment or dementia (n = 27) were matched to a clinical sample (CS) by age, sex, education, and diagnosis (n = 83). ANOVAs compared neuropsychological composites at baseline and over time between retirees and their matched groups. Regression models evaluated whether concussions, concussions with loss of consciousness (LOC), or games played predicted neuropsychological functioning.

**Results:** At baseline, CN retirees had slightly worse memory than controls ($M_{CN\ retirees} = 50.69$, $SE_{CN\ retirees} = 1.320$; $M_{Healthy\ controls} = 57.08$, $SE_{Healthy\ controls} = 1.345$; $p = 0.005$). No other group diferences were observed, and head-injury exposure did not predict neurocognitive performance at baseline or over time.

**Conclusions:** Head-injury exposure was not associated with later-in-life cognition, regardless of cognitive diagnosis. Some retirees may exhibit lower memory scores compared to age-matched peers, though this is of unclear clinical significance.

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Professional football; neuropsychological performance; Head-Injury exposure; cognitive impairment; MCI/Dementia

## Introduction

Scientific interest in the potential impact of American football-related repetitive concussive and sub-concussive hitson cognitive aging has risen dramatically in the last 15 years. Several surveys and retrospective analyses (most based on self-report) suggest the risk of mild cognitive impairment (MCI) and dementia may be higher in former National Football League (NFL) players (1–4), and that former professional players report high rates of cognitive and emotional dysfunction (5). However, investigations examining relationships between head-injury exposure and performance-based neuropsychological functioning have reported mixed findings. Prior investigations have suggested former NFL players may exhibit lower verbal memory (6–11), confrontation naming (6, 9–11), and executive function scores (9–12), though these effects have been generally small, and some investigations have not found differences compared to matched controls (13, 14). Regarding a dose–response relationship between head-injury exposure and neurocognitive abilities, one well-cited study suggested that beginning tackle football prior to age 12 was associated with executive dysfunction later-in-life (15), while others have not found statistically significant relationships between neuropsychological functioning and age of first exposure (16), years of pre-high school football (17), number of concussions (18), or years of football played (18) in former NFL retirees. At the amateur level, survey data have not supported a clear dose–response relationship between age of first exposure to tackle football and poor cognitive or mental health outcomes in former high school players (19, 20). Overall, conclusions in the literature are limited by considerable methodological differences and the absence of longitudinal follow-up (for a detailed review, see Schaffert et al. [21]). The present study sought to address a significant knowledge gap in the literature by examining whether head-injury exposure obtained over an NFL career is associated with greater cognitive impairment or changes over time compared to matched comparison groups.

## Methods

### *National Football League retirees*

NFL retiree data were collected via a prospective longitudinal investigation of aging and cognition among professional contact sport retirees at the University of Texas Southwestern Medical Center (UTSW) and University of Texas Dallas (UTD). Retired NFL players were recruited through presentations at local NFL retiree gatherings, meetings of local NFL Players Association Chapter, local advertising, and/or word of mouth. Retirees could not have a history of stroke, epilepsy, and/or pervasive developmental disorder. There were no

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requirements for NFL players to be cognitively symptomatic or asymptomatic.

Out of 93 former professional contact sport athletes participating in the original prospective study, only those who were age 50 or older, played in the NFL, and had available neuropsychological data were selected for this study (n = 53). Former NFL players underwent a clinical interview, comprehensive neuropsychological testing, neuroimaging, and a neurological exam, as previously described (6). Retirees were deemed to have cognitive impairment (n = 27) or be cognitively normal (n = 26) based on actuarial MCI criteria and clinical consensus. These retirees were then matched to cognitively normal healthy controls or a clinical sample of patients from an outpatient neuropsychology clinic, with the details of the diagnostic procedures and specific groups described in the next sections. Years since retirement from the league ranged from 22 to 50 ($M$ = 35.1, $SD$ = 7.67), and years of education ranged from 14 to 19 ($M$ = 16.3, $SD$ = 0.93). Forty-two were Caucasian and 11 were African American. Twenty-nine players from the initial 53 returned for follow-up neuropsychological evaluation 13 to 59 months later ($M$ = 26.1, $SD$ = 12.76).

Detailed concussion histories were collected via self-report during clinical interview and defined using American Academy of Neurology (1997) criteria (21). Medical conditions and surgical history were also collected through questionnaires and self- and/or informant report during a semi-structured clinical interview by either a neuropsychologist or behavioral neurologist. Total number of NFL games played, games started, and years played were collected through a website of archived statistics (www.pro-football-reference.com). Retirees had a mean of 5.63 concussions (2.24 with loss of consciousness [LOC]), 8.89 years in the NFL, 115.12 games played, and 48.66 games started. Primary positions of the players were lineman (n = 22, 42%), running back (n = 8, 15%), linebacker (n = 7, 13%), safety (n = 7, 13%), cornerback (n = 4, 8%), quarterback (n = 2, 4%), and wide receiver (n = 2, 4%).

### Cognitively Normal (CN) controls

Healthy CN control participants were recruited simultaneously with NFL players from healthy aging studies at UTD and underwent neuropsychological testing and neuroimaging. CN controls (N = 26) were matched as closely as possible to CN retired NFL players by age and education. CN controls had no self-reported history of neurological disease, concussion/traumatic brain injury, or participation in contact sports at the collegiate or professional level. CN participants were aged 52 to 77 ($M$ = 64.88, $SD$ = 7.36), and obtained 12 to 20 years of education ($M$ = 16.23, $SD$ = 2.03). Twenty-four were Caucasian and two were African American. Fifteen CN controls returned for a follow-up visit 8 to 38 months later ($M$ = 18.73, $SD$ = 7.43).

### Clinical archival sample

A clinical sample of well-characterized patients with MCI and dementia was used for comparison against CI retirees. Archived data were obtained from the UT Southwestern Neuropsychology Clinic that consisted of patients referred for evaluation as part of routine clinical care. Participants from this pool were matched to impaired NFL retirees as closely as possible in terms of age, race, and diagnosis on an approximately 3:1 ratio (for the matching procedure, see Supplement 1). Clinical patients (n = 83) ranged in age from 53 to 81 ($M$ = 66.39, $SD$ = 7.69) and had 12 to 20 years of education ($M$ = 16.02, $SD$ = 2.28). Seventy-three were Caucasion, and 10 were African American. Data on demographics, neuropsychological results, medical histories, medications, and diagnosis were collected via each clinical participant's neuropsychological report. None of the participants from the clinical sample were undergoing litigation or had a history of concussion/traumatic brain injury, epilepsy, stroke, bipolar disorder, or psychotic disorder. Repeat neuropsychological evaluations were available for 24 individuals from 10 to 101 months later ($M$ = 26.96, $SD$ = 20.23).

### Neuropsychological measures

All participants underwent a detailed neuropsychological evaluation that included common measures of attention, processing speed, executive functioning, language, and verbal and nonverbal memory. Eleven neuropsychological scores were used: Trail Making A (TMTA) and B (TMTB), Wechsler Adult Intelligence Scale-IV (WAIS-IV) Coding, WAIS-IV Digit Span, Controlled Word Association Test (COWAT), Animal Fluency, Boston Naming Test-2$^{nd}$ edition (BNT), Rey-Osterrieth Complex Figure Test (ROCFT) immediate (IR) and delayed recall (DR), and California Verbal Learning Test-2nd Edition (CVLT-II) total learning and delayed recall (DR). Mitrushina regression-based meta norms (22) were used as the normative data for the TMTA, TMTB, COWAT, Animal Fluency, BNT, and ROCFT. Normative data for the CVLT-II, WAIS-IV Coding, and WAIS-IV Digit Span were obtained from their respective reference manuals. These tests were combined to form three neuropsychological composites (average normative T-scores) in order to reduce the number of analyses: Executive functioning/Attention/Speed (EFAS; [TMTA + TMTB + WAIS-IV Coding + WAIS-IV Digit Span])/4), Language ([COWAT + Animal Fluency + BNT]/3), and Memory ([CVLT-II Total Learning + CVLT-II DR + ROCFT IR + ROCFT DR]/4).

### Diagnoses of MCI and dementia

MCI diagnoses for all participants were determined using adapted Jak and Bondi actuarial MCI criteria (23). Actuarial criteria have been utilized in prior aging NFL retiree research (11) and been shown to improve diagnostic accuracy compared to conventional research methods (23). MCI diagnostic criteria were met when participants displayed impairment (i.e., <16$^{th}$ percentile) on two or more scores within one cognitive domain (i.e., executive functioning/attention/processing speed, language, or memory). Because this study did not utilize a measure of functional activities of daily living, dementia diagnoses for the retired athletes were determined by

consensus with a neurologist (J.H.) and two neuropsychologists (N.D. and M.C.) using neuropsychological data, neurologic exam results, clinical interview information (i.e., impairment interfering in basic or instrumental activities of daily living), and neuroimaging. Overall, 26 retired players were determined to be CN, 22 met criteria for MCI, and 5 had dementia presumed to be due to AD.All of the participants in the clinical sample met criteria for MCI (n = 69) or dementia presumed to be due to AD (n = 14). Additional information on diagnoses can be found in Supplement 1.

### Informed consent and IRB approval

The Longitudinal Study of Aging NFL Players study was approved by the UTD and UTSW IRB and all participants signed informed consent. The clinical sample provided written consent for their archived neuropsychological data and medical history to be used for research purposes at the time of neuropsychological evaluation.

### Data analyses

Univariate Analyses of Variance were used to compare the neuropsychological functioning composites between groups of retired NFL players and their respective control samples (i.e., CN retirees vs. CN controls, and CI retirees vs. CS), with Hochberg's GT2 procedure for post-hoc analyses. To evaluate if scores differed between groups over time, a change score was computed (i.e., Time 2 neuropsychological composite – Time 1 neuropsychological composite) and evaluated in Univariate Analyses of Variance. To evaluate if head-injury exposure was related to worse neuropsychological performance in NFL retirees, linear regression models were used to evaluate whether number of concussions, number of concussions with LOC, or games played predicted scores on the three neuropsychological composites(baseline and change scores). A combined sample of both CN and CI retirees was used in linear regressions to increase the available sample size. *P*-value significance was set at <0.05 and checked against wild bootstrapping confidence intervals (2000 iterations with bias-corrected and accelerated confidence intervals) to determine if results remained significant. All analyses were conducted with IBM SPSS Statistics (version 26). Missing data was handled pairwise for each analysis.

## Results

### Sample characteristics

Sample characteristics can be found in Table 1. ANOVA revealed that the groups were not significantly different in terms of age, $F$ (3, 158) = 1.691, $p$ = 0.171, or education, $F$ (3, 158) = 0.356, $p$ = 0.785, and chi-square analysis did not reveal any significant group differences in race, $X^2$ = 4.297, $p$ = 0.231. MCI subtype (see Supplement 1) did not differ between the CS and CI retirees, $X^2$ = 1.168, $p$ = 0.883. CI and CN retirees did not significantly differ in number of concussions, $t$ (46) = −0.527, $p$ = 0.601, number of concussions with LOC, $t$ (49) = −1.313, $p$ = 0.195, years playing professionally, $t$ (51) = −1.427, $p$ = 0.160, games played, $t$ (49) = −1.536, $p$ = 0.131, games started, $t$ (48) = −0.797, $p$ = 0.429, or age started playing tackle football, $t$ (17) = −0.194, $p$ = 0.849.

### Baseline neuropsychological performance between groups

ANOVA models revealed a significant difference between the four groups in EFAS [F (3, 158) = 10.507, $p$ < 0.001)], Language [F (3, 155) = 10.117, $p$ < 0.001)], and Memory [F (3, 158) = 70.745, $p$ < 0.001)] composites. Post-hoc analyses (Hochberg method) with 2000 iterations of wild bootstrapping with bias-accelerated corrections revealed no significant differences between CN retirees and CN controls, or between CI retires and the CS in EFAS or language composites (Table 2). CN retirees and CN controls significantly differed in memory functioning at baseline, with CN retirees performing in the average range overall, and CN controls performing in the above average range (Table 2).

### Neuropsychological change over time between groups

ANOVAs did not reveal significant differences in neuropsychological change over time between groups in EFAS [F (3, 64) = 0.276, $p$ = 0.842)], Language [F (3, 64) = 1.459, $p$ = 0.234)], or memory [F (3, 64) = 2.003, $p$ = 0.122)]. As shown in Table 3, post-hoc analyses revealed no significant differences between CN retirees and CN controls, or between CI retirees and the CS in any neuropsychological composite.

**Table 1.** Sample characteristics.

| | CN Retiree (n = 27) | CN Control (n = 26) | CI Retiree (n = 26) | Clinical Sample (n = 83) |
|---|---|---|---|---|
| Age, M (SD) | 63.3 (7.4) | 64.9 (7.3) | 67.5 (7.5) | 66.4 (7.7) |
| Education, M (SD) | 16.4 (1.1) | 16.2 (2.0) | 16.2 (0.7) | 16.0 (2.3) |
| AfricanAmerican, N (%) | 7 (25.9) | 2 (7.7) | 4 (15.4) | 10 (12.0) |
| Concussions, M (SD) | 5.3 (4.1) | – | 6.0 (5.0) | – |
| Concussions w/LOC, M (SD) | 1.8 (1.8) | – | 2.7 (2.7) | – |
| Games played, M (SD) | 104.7 (53.9) | – | 126.0 (44.4) | – |
| Games started, M (SD) | 42.9 (45.8) | – | 54.9 (59.8) | – |
| NFL Years Played, M (SD) | 8.2 (3.8) | – | 9.6 (3.1) | – |

Note: LOC = loss of consciousness; NFL = National Football League; CN = Cognitively Normal; CI = Cognitively Impaired

**Table 2.** Baseline neuropsychological performance (T-scores) between groups.

| | | | | | Bootstrap for Mean | |
|---|---|---|---|---|---|---|
| Composite | Group | $M_{(T\text{-}scores)}$ | 95% CI | $p$ | Bias | 95% CI |
| EFAS | CN Retirees | 51.69 | 48.93, 54.44 | 0.835 | −0.036 | 49.96, 53.41 |
| | CN Controls | 53.93 | 51.12, 56.73 | | 0.040 | 52.14, 55.87 |
| | CI Retirees | 47.52 | 44,71, 50.32 | 0.882 | 0.015 | 44.53, 50.57 |
| | CS | 45.83 | 44.26, 47.40 | | 0.199 | 43.73, 48.37 |
| Language | CN Retirees | 50.53 | 47.76, 53.30 | 0.548 | 0.024 | 48.03, 53.07 |
| | CN Controls | 53.71 | 50.71, 56.71 | | −0.007 | 51.38, 55.94 |
| | CI Retirees | 44.05 | 41.23, 46.87 | 0.760 | 0.007 | 41.45, 46.78 |
| | CS | 46.09 | 44.51, 47.67 | | 0.112 | 44.16, 48.18 |
| Memory | CN Retirees | 50.69 | 48.08, 53.29 | 0.005* | −0.036 | 48.33, 52.84 |
| | CN Controls | 57.08 | 54.42, 59.74 | | 0.031 | 54.29, 59.88 |
| | CI Retirees | 38.01 | 35.35, 40.67 | 0.999 | 0.001 | 35.15, 40.88 |
| | CS | 37.41 | 35.93, 38.90 | | 0.116 | 35.77, 39.33 |

Note: EFAS = Executive Functioning/Attention/Speed; CN = Cognitively Normal, CI = Cognitively Impaired, CS = Clinical Sample
*Denotes significance at $p < 0.05$ with 2000 iterations of wild bootstrapping between CN retirees vs. CN controls, or CI retirees vs. CS

**Table 3.** Neuropsychological change over time between groups.

| | | | | | Bootstrap for Mean | |
|---|---|---|---|---|---|---|
| Composite | Group | $M_{change\ (T\text{-}scores)}$ | 95% CI | $p$ | Bias | 95% CI |
| EFAS | CN Retirees | −0.27 | −2.42, 1.88 | 0.999 | −0.03 | −2.08, 1.55 |
| | CN Controls | −0.90 | −3.05, 1.26 | | −0.01 | −2.46, 0.60 |
| | CI Retirees | −1.65 | −3.88, 0.59 | 0.985 | 0.04 | −4.15, 0.94 |
| | CS | −0.72 | −2.42, 0.98 | | −0.05 | −2.26, 0.72 |
| Language | CN Retirees | 0.73 | −1.56, 3.01 | 0.999 | −0.04 | −1.41, 2.74 |
| | CN Controls | 0.36 | −1.92, 2.65 | | −0.01 | −0.98, 1.70 |
| | CI Retirees | −2.44 | −4.80, −0.07 | 0.685 | 0.07 | −4.99, 0.25 |
| | CS | −0.42 | −2.22, 1.39 | | −0.40 | −2.42, 0.79 |
| Memory | CN Retirees | −1.01 | −4.02, 1.85 | 0.999 | −0.04 | −3.70, 1.42 |
| | CN Controls | −1.82 | −4.75, 1.11 | | 0.03 | −4.24, 0.68 |
| | CI Retirees | −2.46 | −5.50, 0.57 | 0.194 | −0.04 | −6.10, 1.07 |
| | CS | 1.63 | −0.68, 3.96 | | −0.29 | −0.37, 3.13 |

Note: EFAS = Executive Functioning/Attention/Speed; CN = Cognitively Normal, CI = Cognitively Impaired, CS = Clinical Sample
*Denotes significance at $p < 0.05$ with 2000 iterations of wild bootstrapping between CN retirees vs. CN controls, or CI retirees vs. CS

### *Neuropsychological performance at baseline and over time predicted by head-injury exposure*

Multiple regressions were used to evaluate whether EFAS, language, or memory composites were predicted by number of concussions, number of concussions with LOC, or NFL games played. None of the regression models reached statistical significance. Baseline functioning in EFAS ($R^2$ = 0.013, F [3,42] = 0.186, $p$ = 0.905), language ($R^2$ = 0.098, F [3,43] = 1.566, $p$ = 0.211), or memory ($R^2$ = 0.129, F [3, 43] = 2.154, $p$ = 0.111) were not significantly predicted by number of concussions, number of concussions with LOC, or NFL games played. Similarly, change over time in EFAS ($R^2$ = 0.069, F [3,24] = 0.589, $p$ = 0.628), language ($R^2$ = 0.078, F [3, 24] = 0.674, $p$ = 0.577), or memory ($R^2$ = 0.012, F [3, 24] = 0.094, $p$ = 0.963) were not significantly predicted by number of concussions, number of concussions with LOC, or NFL games played.

### *Secondary analysis: speed vs. non-speed players and baseline neurocognitive performance*

To assess the impact of football position played on neurocognitive outcomes, post-hoc analyses were conducted by stratifying players into non-speed (n = 30; quarterback, lineman, linebacker, special teams) and speed (n = 23; running back, cornerback, wide receiver) positions. There were 14 CN retirees (matched to 13 CN controls) and 16 CI retirees (matched to 48 CS controls) in the non-speed group, and 13 CN retirees (matched to 12 CN controls) and 10 CI retirees (matched to 35 CS controls) in the speed group.

In both non-speed and speed players bootstrapped t-test examining group differences stratified by position (non-speed vs. speed) revealed no significant differences between CN retirees and their controls, or in CI retirees and the CS, in baseline EFAS or language. Similar to the main analyses, baseline memory performance was slightly higher for CN controls, compared to CN retirees, though this did not appear to be driven by position (Table 4 and Table 5). We did not evaluate change over time due to limited sample size.

In non-speed players, head-injury exposure variables did not significantly predict baseline EFAS ($R^2$ = 0.239, F [3, 21] = 2.199, $p$ = 0.118), language ($R^2$ = 0.254, F [3, 21] = 2.381, $p$ = 0.098), or memory ($R^2$ = 0.236, F [3, 21] = 2.157, $p$ = 0.123) performance. Similarly in speed players, head-injury exposure variables did not significantly predict baseline EFAS ($R^2$ = 0.082, F [3, 18] = 0.539, $p$ = 0.661), language ($R^2$ = 0.005, F [3, 18] = 0.033,

**Table 4.** Baseline neuropsychological performance (T-scores) between groups (non-speed players only).

| | | | | | Bootstrap for Mean | |
|---|---|---|---|---|---|---|
| Composite | Group | $M_{(T\text{-}scores)}$ | $t$ | $p$ | Bias | 95% CI |
| EFAS | CN retirees | 52.80 | −0.685 | 0.500 | −0.025 | 50.31, 55.17 |
| | CN Controls | 54.10 | | | 0.042 | 51.24, 56.97 |
| | CI retirees | 46.42 | 0.103 | 0.919 | −0.029 | 42.21, 50.48 |
| | CS | 46.17 | | | −0.061 | 43.91, 48.29 |
| Language | CN retirees | 51.18 | −1.903 | 0.070 | −0.001 | 47.99, 54.39 |
| | CN Controls | 55.72 | | | −0.027 | 52.96, 58.45 |
| | CI retirees | 45.03 | −0.274 | 0.785 | −0.008 | 41.12, 48.46 |
| | CS | 45.71 | | | −0.007 | 43.12, 48.25 |
| Memory | CN retirees | 51.22 | −2.521 | 0.018* | −0.021 | 47.97, 54.80 |
| | CN Controls | 58.34 | | | −0.028 | 47.25, 53.25 |
| | CI retirees | 37.96 | 0.096 | 0.923 | −0.021 | 33.89, 42.28 |
| | CS | 37.76 | | | 0.019 | 35.87, 39.71 |

Note: EFAS = Executive Functioning/Attention/Speed; CN = Cognitively Normal, CI = Cognitively Impaired, CS = Clinical Sample
*Denotes significance at $p < 0.05$ with 2000 iterations of bootstrapping between CN retirees vs. CN controls, or CI retirees vs. CS

**Table 5.** Baseline neuropsychological performance (T-scores) between groups (speed players only).

| | | | | | Bootstrap for Mean | |
|---|---|---|---|---|---|---|
| Composite | Group | $M_{(T\text{-}scores)}$ | $t$ | $p$ | Bias | 95% CI |
| EFAS | CN retirees | 50.49 | −1.671 | 0.108 | 0.053 | 47.56, 53.26 |
| | CN Controls | 53.75 | | | 0.022 | 51.15, 56.49 |
| | CI retirees | 49.27 | 1.362 | 0.180 | 0.043 | 43.94, 53.99 |
| | CS | 45.38 | | | 0.003 | 42.74, 47.94 |
| Language | CN retirees | 49.82 | −0.775 | 0.446 | 0.051 | 46.17, 53.79 |
| | CN Controls | 51.87 | | | 0.057 | 48.43, 55.81 |
| | CI retirees | 42.48 | −1.806 | 0.078 | 0.034 | 38.49, 46.54 |
| | CS | 46.59 | | | 0.026 | 44.28, 48.74 |
| Memory | CN retirees | 50.10 | −2.276 | 0.032* | −0.009 | 48.25, 43.25 |
| | CN Controls | 55.83 | | | 0.002 | 52.19, 59.45 |
| | CI retirees | 38.08 | 0.508 | 0.614 | −0.034 | 34.22, 41.53 |
| | CS | 36.97 | | | −0.002 | 34.96, 39.12 |

Note: EFAS = Executive Functioning/Attention/Speed; CN = Cognitively Normal, CI = Cognitively Impaired, CS = Clinical Sample
*Denotes significance at $p < 0.05$ with 2000 iterations of bootstrapping between CN retirees vs. CN controls, or CI retirees vs. CS

$p$ = 0.992), or memory ($R^2$ = 0.076, F [3, 18] = 0.494, $p$ = 0.691) performance. We did not evaluate change over time due to limited sample size.

## Discussion

This is the first study to investigate the neuropsychological functioning of retired NFL athletes longitudinally, and one of only three studies to compare the neuropsychological functioning in NFL athletes with MCI and dementia to a clinical sample (3,24,25). Our findings revealed that the neuropsychological functioning of NFL retirees did not differ from comparison groups on most neuropsychological composites. Only baseline memory performance was significantly lower in CN retirees compared to CN controls, though CN retirees' performance fell solidly within the average range and the mean performance was less than 1 SD apart in normative score difference. We also analyzed dose–response relationships between neuropsychological performance and number of concussions, concussions with loss of consciousness, and NFL games played, and found that these head-injury exposure variables did not predict neuropsychological functioning later-in-life, or neuropsychological change over time. Furthermore, we did not find that player position impacted these findings.

Differences in memory among retired NFL players compared to age-matched healthy controls have been previously reported (9,22). In the current study, we found CN retirees had slightly lower memory scores compared to healthy controls but did not find it to be significantly associated with head-injury exposure. Previous research using this cohort found hippocampal volume, a region intimately tied to memory functioning, to be reduced in players with NFL careers of over 120 games and with a greater frequency of concussions with LOC (7). However, with a larger sample size in the current study, we did not find memory performance to be related to number of concussions with LOC. Notably, we also only found a difference between CN retirees and a healthy control group, and not in former CI players compared to a clinical sample. In addition, despite a statistically significant result, it is important to emphasize that CN controls performed in the above average range ($M_{\text{Memory Composite T-Score}}$ = 57.08) which is at the 77$^{th}$ percentile for their age, and as a group, the CN retirees memory still fell solidly within the average range ($M_{\text{Memory Composite T-Score}}$ = 50.69), suggesting that the group difference was driven by sampling an above-average healthy control group rather than a poorer functioning NFL cohort. Additionally, we did not find that these group differences varied by position played, with a similar effect observed in both speed and non-speed

positions. As such, our findings would suggest little to no detectable impact of cumulative head injury exposure on memory ability, though future studies are needed to validate this finding given our small sample.

This is one of the first investigations to compare the neuropsychological functioning of cognitively impaired former NFL players to samples of clinical patients with the same diagnoses, and only two previous studies to date have utilized a similar design. A recently published pilot investigation using a smaller sample of this cohort (N = 10 retirees with MCI and N = 10 matched clinical patients with MCI) found differences in confrontation naming, visual speed, and divided attention which were not replicated by the current findings (24, 26). This could be due to the use of a larger sample, different normative procedures for most measures (e.g., regression based), the use of neuropsychological composites, and different diagnostic procedures (actuarial criteria versus conventional clinical MCI criteria) in the current study. Nonetheless, our larger sample size and more powerful analyses did not yield significant differences regarding EFAS or language functioning. The only other similar evaluation of former players with MCI was published in 2013, where Randolph et al. (3) compared performance on the Repeatable Battery for the Assessment of Neuropsychological Status (RBANS) between retired NFL players with amnestic MCI and clinical controls with amnestic MCI recruited from academic medical centers. No significant differences were seen between NFL retirees and the clinical sample in RBANS Index scores measuring episodic memory, visuospatial abilities, language, or attention. The authors also did not find a significant association between years of professional play and scores on the RBANS Total Score Index (3). Results from the current investigation generally echo these findings.

Some have posited that cumulative "subconcussive" head impacts over a long NFL career may reduce cerebral reserve and lead to an earlier expression of cognitive impairment later in life (27), or even be associated with risk of CTE (28). In the current study, we expanded upon recent work utilizing this cohort (18) with a significantly larger sample, more robust statistical analyses, and different head-injury exposure variables examined. We used games played as a proxy measure for "subconcussive" impacts, and we did not find any association between greater head-injury exposure and neuropsychological functioning. In addition to findings from this cohort of retired NFL players, investigations of other cohorts have failed to find an association between more football exposure and neuropsychological performance. Perhaps the most comprehensive study to date, Solomon and colleagues (17) used an extensive clinical neuropsychological battery (nine tests measuring verbal and visual memory, processing speed, attention, language, and executive functioning) and found no association between cognitive performance and years of pre-high school football after correcting for multiple comparisons. However, it should be noted that a recent publication may suggest that use of a comprehensive head-injury composite that takes into account practice time and helmet impact data may be more sensitive to cognitive functioning in former NCAA football players (29). While we did not have all the necessary data in this cohort to use such a composite, our findings did not reveal any significant dose–response relationships among these 53 former NFL players. Future studies that carefully investigate longitudinal neuropsychological functioning with more sensitive head-injury exposure variables are needed in this regard.

Although the current study has several methodological strengths and is the first longitudinal evaluation of neuropsychological performance in an older group of retired NFL players, it does have several limitations. First, our clinical sample of patients with MCI and dementia were not recruited prospectively alongside our retired NFL players. Although we attempted to match this sample by demographics and diagnosis, other group differences and potentially confounding factors could exist because we were limited to collecting medical and background information contained inneuropsychological reports. Information contained in neuropsychological reports can vary between clinicians, and it is possible that some background information could have been left out of the report if it was not perceived as relevant to the clinical referral question. Second, although we made sure that all participants with MCI met the same actuarial criteria, clinical patients were initially screened and collected based on individual diagnoses among various clinicians, possibly introducing some sampling noise. It is also important to note that neither the NFL nor control samples had available standardized functional measures to differentiate MCI and dementia; thus, it is possible that some participants' diagnoses may have changed if these measures were included. Third, although our CN control group was recruited alongside our NFL sample, they exhibited above average memory performance, and may not represent the general population of cognitive normal aging individuals. Finally, we collected concussion data via self-and/or informant report, and, as with most concussion research, corroboration with medical records was not feasible, and thus some of these data may be prone to recall bias. However, each player was carefully interviewed and asked to describe their concussion history in detail using a semi-structured interview pertaining to concussion history throughout their playing career, post-traumatic amnesia, lingering symptoms, and recovery timelines, giving us more confidence in the validity of head-injury exposure levels within our sample.

Overall, our results suggest that some cognitively normal NFL retirees may perform at a lower level (yet within normal limits) on memory measures, though the clinical significance of this finding is unclear. However, we did not find evidence that NFL retirees had disproportionate cognitive decline over time compared to non-NFL participants, regardless of the presence or absence of cognitive impairment. Furthermore, the current study did not find evidence that head-injury exposure obtained over an NFL career is related to changes in neurocognitive ability later-in-life. Future, larger prospective longitudinal studies examining the neuropsychological functioning of NFL retirees over time are needed to help clarify risk of long-term cognitive impairment related to American football participation.

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## Disclosure statement

Munro Cullum is currently serving as the Scientific Director of the Texas Alzheimer's Research and Care Consortium. All other authors report no conflicts of interest.

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